
CSIRO PUBLISHING

Australian Journal of Soil Research



Volume 38, 2000
© CSIRO Australia 2000

A journal for the publication of original research
in all branches of soil science

www.publish.csiro.au/journals/ajsr

All enquiries and manuscripts should be directed to

Australian Journal of Soil Research

CSIRO PUBLISHING

PO Box 1139 (150 Oxford St)

Collingwood

Vic. 3066

Australia

Telephone: 61 3 9662 7628

Facsimile: 61 3 9662 7611

Email: jenny.fegent@publish.csiro.au



Published by **CSIRO PUBLISHING**
for CSIRO Australia and
the Australian Academy of Science



Particle density and the plummet balance

W. W. Emerson and D. Weissmann

CSIRO Land and Water, PMB No. 2, Glen Osmond, SA 5064, Australia.

Abstract

Soil suspensions are usually characterised by the percentage weights of particles settling with velocities between defined limits, such as silt and clay. If percentages were measured sufficiently accurately by both the pipette method and a plummet balance, dry densities of fractions could be calculated and hence their composition inferred. The accuracy and precision of the pipette method were confirmed by the mass balance of particles in peptised suspensions of a red-brown earth subsoil and a kaolinite. Percentages indicated by a plummet balance were corrected for the actual densities of fractions measured separately. Residual differences from the pipette values showed that the balance was accurate but lacked the precision needed to deduce densities to 0.01 mg/mm³.

Values of the density of clay fractions were deduced from the means of published percentages obtained by the two methods where replicate suspensions were used and also from the mean difference based on a comparison using 199 soil samples. For the latter, an underestimate of clay percentages indicated by the balance had first to be allowed for. Clay-deduced densities were in the range 2.80–2.83 mg/mm³, compared with 2.84 mg/mm³ measured on the separated subsoil clay. The values are similar, probably because the clays are derived from mica. The density of the separated subsoil silt was 2.64 mg/mm³, less than quartz, because of the presence of feldspar. Such results indicate the potential use of the method if the precision of the balance is improved.

Additional keywords: particle size, iron oxide, tailings, hydrometer.

Introduction

Accurate values of the dry densities of fine fractions of particles in soils or tailings from mines could be useful indicators of both the composition and settlement rates of particles. However, separation of fractions by their velocity of settlement prior to determining dry density is a tedious process. A peptised suspension has to be made, followed by repeated decantation of particles settling with less than a certain velocity. The dilute suspension of clay obtained, for example, then has to be flocculated and concentrated, and the salt used for flocculation washed out if possible, followed by filtering and drying. Finally density has to be measured accurately. However, the first step, preparation of a peptised suspension, is part of the normal process for determining the percentages by weight of clay and silt present. Usually, after the suspension has stood for a given time, either the weight concentration of particles or the suspension density is measured at a depth appropriate to the maximum settling velocity of the fraction. For the weight concentration, a pipette sample is taken and oven-dried and for suspension density either a hydrometer or a plummet balance calibrated for a given particle density (ρ_c) is used. If both values are determined, then the dry density of a fraction (ρ_s) can be deduced (see Appendix, Eqn 2). The values will only be the same if $\rho_c = \rho_s$.

Hutton (1955) tested the balance designed by Marshall (1956) against the pipette method using a polythene plummet. The balance was calibrated for particles of nominal density 2.61 mg/mm³ because of the ISSS (1934) definitions of silt and clay. These

specified not only the diameters of the largest silt and clay particles ($2 \mu\text{m}$, $20 \mu\text{m}$) but also their rates of settlement in water at 20°C . Hutton (1955) used Stokes Law to calculate the density of spherical particles which satisfied the definitions. Plummets calibrated in this way became widely used in Australia to measure silt and clay percentages (McIntyre and Loveday 1974). Hutton (1955) also decanted the silt plus clay from the test suspensions and weighed the sand left. The sum of the clay plus silt measured by the plummet and the sand was found to be often greater than the initial weight of soil taken, showing that $\rho_s > \rho_c$. Hutton (1955) also applied both methods to suspensions of a series of soils whose clay fractions consisted of kaolin with a range of iron oxide contents. The density of the clay with low oxide content deduced using Eqn 2 was found to be much less than that of kaolinite.

Day (1953) compared the pipette and hydrometer methods on suspensions of 13 soils. He found that better agreement between percentages of silt and clay would have been obtained if the hydrometer had been calibrated for particles of density 2.75 mg/mm^3 rather than the standard value of 2.65 mg/mm^3 (quartz). The higher value was used in two later comparisons of the plummet balance and pipette methods (Slager and Koenigs 1964; Sheldrick and McKeague 1976). The authors found the two methods in approximate agreement. However, apart from Hutton (1955), little attempt has been made to relate the usually small difference between percentages determined by the two methods to the density and composition of fractions.

The pipette method and a plummet balance have been applied here to suspensions of samples of a red-brown earth subsoil and a mined kaolinite. These materials were chosen for the contrasting composition and density of their clay fractions. Both samples can also be completely dispersed with polyphosphate, so no material is lost in preparing suspensions. The plummet balance used was calibrated after Hutton (1955), to allow comparison with his test data. After taking the measurements, clay and silt fractions were decanted off and dried, and their densities measured accurately. Differences in density were then related to the composition of the fractions and also used to correct the balance readings. If the pipette method was accurate, then the residual difference between the two methods would indicate the precision of the balance. This precision would be compared with that needed to detect significant differences in fraction composition.

Experimental procedures

Samples used

The red-brown earth subsoil selected was an archival $<2 \text{ mm}$ sample taken by Litchfield (1951) from the B_1 -horizon of Urrbrae fine sandy loam at a depth of $0.33\text{--}0.76 \text{ m}$ (CSIRO Ref. No. 12723; Merry 1985). The profile is also classified as a red chromosol (Isbell 1996) or a calcic rhodoxeralf. Litchfield (1951) reported that the sample had a clay content of 65%, a pH (1:5, H_2O) of 6.6, and contained 0.74% organic C, 5.6% free iron oxide (expressed as Fe_2O_3), and 0.01% NaCl. The clay minerals present consist mainly of illite and kaolin in the approximate ratio 7/3 with a total surface area of $250 \text{ m}^2/\text{g}$ (Burford *et al.* 1962).

The sample of kaolinite was an 'as mined' china clay from St Austell, Cornwall. A small amount of mica was reported to be present besides kaolinite (R. Bryant, pers. comm.). The sample contained $<0.01\%$ total soluble salts. A subsample was lightly crushed to pass a 2-mm sieve.

Apparatus

The general design of the plummet balance is given in Marshall (1956) and notes on its practical operation in McIntyre and Loveday (1974). The plummet following Hutton (1955) is made from polythene rod 25 mm in diameter with an overall volume of 35 mL . The main part is 530 mm long. The top is parabolic in shape with a small flat into which a hook is screwed, while the base is hemispherical. As used here, the plummet was suspended by a nylon thread 0.12 mm in diameter, with only a dot on it to

indicate the required depth of immersion. This eliminated any jerkiness in the balance reading caused by changes in the surface tension on the thread at the air–water interface.

The scale of the balance is divided into 1% intervals and can be read to 0.1–0.2%. The balance was set to read 0% at 20°C with the plummet immersed in water and 100% when immersed in a solution containing 17.47 g/L of NaCl (Hutton 1955). At 20°C, using linear extrapolation of published values (ICT 1932) the solution has a relative density of 1.01053 mg/mm³. From Eqn 1 (see Appendix), this is the same as that of water of relative density 0.99823 mg/mm³ containing 20 g/L of particles of density 2.593 mg/mm³. The particle density calculated from the ISSS (1934) definitions of the settling velocities of the largest clay and silt sized particles is 2.596 mg/mm³ (McIntyre 1974).

The pipette, which was used for taking samples, was connected to a separate hand-operated rack and pinion mounted on a tripod. A pipette filler was attached before lowering it to the required depth in the suspension and sampling.

Preparation of soil suspensions

Ten mL of a 10% solution of calgon, 5 mL of 1 M NaOH, and water were added to duplicate 25-g samples of air-dry kaolinite contained in 600-mL beakers. Water (250 mL) was added and the samples stirred at high speed for 20 min using a motor dispersion unit (Hutton 1955; McIntyre and Loveday 1974). Soil (25 g) which had been oven-dried and then cooled in a desiccator containing P₂O₅ was treated in the same way. The same amounts of calgon and alkali were also added to 25 g of air-dry soil in a 250-mL plastic bottle, followed by 100 mL of water. The bottle was then shaken end-over-end for 64 h. The resultant suspensions were washed through a 70 mesh sieve into a measuring cylinder and diluted with water to bring the volume to 1250 mL.

Pipette samples and plummet balance readings

Suspensions were stirred and then readings taken after 9 min 36 s and 16 h. The balance reading was taken with the mid-point of the plummet at a depth of 205 mm below the liquid surface and then a 5-mL pipette sample at 200 mm. The difference in depth was to allow for the upward displacement of the suspension by the plummet (Marshall 1956). The suspension was re-stirred between readings for the clay plus silt. For clay, the order of sampling was found to be immaterial. Pipette samples were oven-dried and weighed to 0.1 mg. Plummet balance readings were corrected for the blank containing dispersant only.

Concentrating decanted clay and silt

First clay and then silt were exhaustively decanted from the suspension made from air-dried soil. The large volume of suspended clay was flocculated by adding solid NaCl to bring the concentration to 1 M. The supernatant was then decanted and replaced with water to bring the concentration in suspension to 0.25 M. One portion was then poured onto a 0.2- μ m millipore filter and a vacuum of about 700 mm Hg applied. The filter cake was weighed and left to air-dry. The loss in weight was measured so that the amount of NaCl remaining could be calculated approximately. Another portion was washed with 0.1 M CaCl₂. The CaCl₂ concentration was then reduced to 0.001 M by dialysis before filtering and drying. The decanted silt was then allowed to settle, filtered, and then air-dried. Subsamples were subsequently oven-dried. The residue of sand was dried and weighed to check the mass balance.

Measurements of density of fractions

Duplicate measurements of the density were made using water displacement. The temperature in the room during subsequent determinations of particle density varied between 19.8°C and 21.3°C. So with a 100-mL density bottle and only 2–5 g of clay, the change with temperature in the volume of the bottle and the density of water was taken into account (Lambe 1951).

In the case of the clay fraction of the soil in the Na-form, spontaneous dispersion of the clay in the added water had to be avoided. The weighed clay in the density bottle was exposed initially to water vapour. The bottle plus clay was then placed on a laboratory jack and a tube inserted which was connected to a separating funnel filled with water. The rate of flow of water through the tube was controlled by using a Hofmann clip. Water was dripped in slowly at first. Once a free water surface was established, the bottle was gradually raised so as to keep the tip of the tube just below the water surface.

Pre-wetting in the vapour phase did not prevent some air becoming entrapped in the kaolinite even when water was run slowly in. Hence, a 10-g air-dry sample was vigorously shaken with 25 mL of H₂O to form a smooth paste. This was then poured into the density bottle and the bottle filled as before.

Tongs were used both to insert the stoppers and to transfer the bottles for weighing. The temperature of the supernatant was measured after each weighing. Values of density obtained were converted to an oven-dry basis using the air-dry water contents of the samples.

Results

Densities and composition of separated silt and clay

Densities quoted are the mean of duplicates which differed by not more than 0.004 mg/mm^3 . On an oven-dry basis, the density of the Ca-clay separated from the subsoil sample was 2.800 mg/mm^3 . The value for the clay as dried from 0.25 M NaCl was 2.843 mg/mm^3 . The percentage by weight of NaCl present was calculated to be 1.1%. The effect on the measured density would be negligible as it is equivalent to adding a solid of the same weight with a density of 2.86 mg/mm^3 (Emerson and Self 1994). The difference in density between the two forms of the clay means an increased water retention of the Ca-clay of 0.77% by weight.

An approximate density of the Ca-clay can be calculated from its known composition. Fine grained Ca-illite, e.g. Willalooka (Emerson 1964), has a density of 2.75 mg/mm^3 and soil kaolin about 2.61 mg/mm^3 . Therefore, a 70/30 mixture would give a density of 2.71 mg/mm^3 . Deshpande *et al.* (1968) reported that the clay fraction of a sample from the same site taken at 0.35–0.50 m contained 7.1% free iron oxide. Assuming a density of 4.4 mg/mm^3 (goethite) for the iron oxide, the clay fraction would have a density of 2.78 mg/mm^3 , as compared with the measured value of 2.80 mg/mm^3 .

The separated silt from the subsoil had a density of 2.642 mg/mm^3 on an oven-dry basis. Analysis by X-ray diffraction showed that, apart from the quartz, which is dominant, about 5% of albite is present with a lesser amount of sanidine, but no iron oxide. The feldspars found in the subsoil have densities of 2.62 and 2.56 mg/mm^3 , respectively (Huang 1977). Their presence would account for the density of the silt being less than that of quartz.

The oven-dry density of the kaolinite was found to be 2.610 mg/mm^3 , the same as its crystallographic density. It was presumed, therefore, that both silt and clay fractions have the same composition and density.

Accuracy of the pipette method and comparison with the plummet balance

As the oven-dried pipette samples were weighed to 0.1 mg, a precision of better than 0.2% should be obtained for the clay and silt plus clay percentages. For the initially air-dry subsoil, the sum of the percentage by weight of fractions, including sand, came to 99.8%. A small apparent loss in weight would be expected from the measured differences in density of the oven-dried Ca- and Na-forms of the clay. If all the clay were present in the Ca-form initially, its conversion to the Na form by the added dispersant would result in a loss of 0.5% (62% of 0.77%).

The percentage by weight of clay present in the subsoil as deduced from the reading on the plummet balance minus the reading in the blank was appreciably greater than that found by the pipette method (Table 1). The correction factor from Eqn 2 (Appendix) for the difference between the density used in calibrating the balance and the measured density of 2.84 mg/mm^3 is 0.948. When this was applied there was approximate agreement (see Table 1). The silt percentages given for the plummet balance were deduced directly from the differences between the scale readings for clay plus silt and clay particles. Agreement between the plummet balance and pipette values was again improved when the correction factor (0.988) for a density of 2.64 mg/mm^3 was applied (Table 1).

Table 1. The percentage weights of fractions on an oven-dry basis measured gravimetrically and by a plummet balance calibrated for particles of density 2.593 mg/mm³ before and after correcting for the measured average particle density (ρ_s) of each fraction

Fraction	Pipette method (%)	Plummet balance (%)	ρ_s (mg/mm ³)	Correct for ρ_s	Error (%)
<i>Subsoil</i>					
Clay	62.8	66.4	2.84	63.0	+0.2
Air-dry	63.1	66.2	„	62.8	-0.3
Oven-dry					
Silt	16.9	17.5	2.64	17.3	+0.4
Air-dry	15.7	16.1	„	15.9	+0.2
Oven-dry					
<i>Kaolinite</i>					
Clay	53.5	53.6	2.61	53.4	-0.1
	50.9	51.5	„	51.3	+0.4
Clay +	98.6	98.6	„	98.2	-0.4
silt	100	100.4	„	100	0

The percentages by weight of silt plus clay in the two soil suspensions as measured with the pipette method were 79.7 and 78.8. The percentages indicated on the balance exceeded the pipette values by 4.2 and 3.5, respectively (Table 1), and correspond to 5.0% and 4.5% of the clay plus silt, respectively. Previous unpublished results (W. W. Emerson and D. Weismann) for samples from the B₂ horizons of 3 other red-brown subsoils around Adelaide showed that the weight percentages indicated by the balance for silt plus clay had to be reduced by 4.4%, 5.2%, and 5.3%, respectively, to make their sum, together with the sand, equal to 100%.

Clay and silt plus clay percentages in the kaolinite suspensions measured by pipette and indicated on the balance after subtracting the reading in the blank are shown in Table 1. There was a small difference between the duplicate suspensions of kaolinite. The correction factor for the difference between the measured density of the fraction and the calibration density is 0.996. As shown in Table 1, the correction although small has reduced the average difference from +0.25% to zero.

Fraction densities calculated from published comparisons of the pipette method and plummet balance

Slager and Koenigs (1964) used 9 replicate 1% suspensions of the oven-dried, fine earth fraction of Winsum clay loam dispersed using polyphosphate. The plummet in this case was suspended from a torsion balance. The average clay content measured by pipette was 25.0% and by the balance, 25.4%. Using Eqn 2, the average particle density of the clay was, therefore, 2.83 mg/mm³. The average silt plus clay content was the same by both methods (39.0%). The density of the silt fraction was therefore about 2.69 mg/mm³. Using a plummet balance as used in this research, Sheldrick and McKeague (1976) reported duplicate measurements of clay percentages on one soil. The values were 27.8% by plummet and 27.45% by pipette, indicating a clay density of about 2.8 mg/mm³. Both values for the clay are similar to that measured here on the red-brown earth subsoil.

Hutton (1955) used 199 soil samples taken during soil surveys mainly of irrigation areas in southern Australia. The balance was found to indicate about the same percentage

of clay as the pipette method. The average density of the clay was therefore apparently equal to the calibration density of 2.593 mg/mm^3 . However, the percentage weight of silt plus clay indicated was over-estimated. When the percentage was added to the percentage weight of sand present, the sum often exceeded the initial weight of the sample. Hutton (1955) found that if the indicated percentage of silt plus clay was reduced by 5%, the sum for most soils came to $99\% \pm 2\%$ of the weight of the initial sample. This reduction is similar to that found in this research, for the test subsoil and previously for the 3 other red-brown earth subsoils.

Hutton (1955) also compared the two methods on 10 soils whose clay fraction consisted of kaolin and iron oxide only. The density of each clay fraction was found by the value required to convert the clay percentage indicated by the balance to that found by pipette. Extrapolation to zero iron oxide content gives a value of about 2.44 mg/mm^3 for the density of soil kaolin, in contrast to the known density of kaolinite, 2.61 mg/mm^3 .

Discussion

Density of clay fractions and apparent errors in using the plummet balance

Densities of $2.80\text{--}2.84 \text{ mg/mm}^3$ seem to be common values for soil clay fractions, apart from the much smaller average value deduced from the results of Hutton (1955). The latter result was puzzling, because many of the samples would have come from red-brown earth profiles. Radoslovich (1958) showed that the clay mineral composition of such profiles in Southern Australia is similar, probably because mica is dominant in the schists, shales, and slates from which the soils are derived (Norrish and Pickering 1983). The results of Litchfield (1951) for the type profile also show that at least the total free iron oxide present down the profile is a constant fraction of the clay content. The reduction in the percentage of silt plus clay suggested by Hutton (1955) to preserve the mass balance of the plummet balance method is also similar to that found in this research both for the test red-brown earth sample and previous samples from similar profiles. Further, it would be expected that the density of kaolin should decrease to that of kaolinite as the proportion of free iron oxide present decreases and presumably, with it, any iron substitution in the clay lattice. Fortunately, Marshall (1956) noted that 'In an alternative method of immersion used by Hutton (1955), the plummet throughout remains attached to the balance'. This is interpreted to mean that after inserting the plummet to 25 cm and taking the silt plus clay reading it was left in the suspension. During the 18 h before reading the clay percentage on the balance, fine particles would have settled on the plummet causing a *reduction* in the reading. Using Eqn 2, to bring the density for kaolin up from 2.44 mg/mm^3 to the 2.61 mg/mm^3 of kaolinite would require that the ratio of the true clay percentage to that measured was $(1 - 1/2.44)/(1 - 1/2.61)$ or 1.046. For the tests on 199 samples, the same increase would result in an average density of 2.80 mg/mm^3 [$1.046 = (1 - 1/2.593)/(1 - 1/2.80)$]. So the error in the mass balance in the tests on soils is mainly accounted for by the high densities of the clay fraction. The silt will have a density near quartz, so that the total over-estimate of silt plus clay is about the 5% suggested by Hutton (1955) to correct the mass balance. Sediment collection on the plummet probably was not detected because the fine particles washed off as the plummet was quickly raised out of the suspension. In the subsequent routine version of the method, the plummet was withdrawn after taking the silt plus clay reading and re-inserted to measure clay. No corrections were applied for density. In the course of these measurements, H. J. Beatty (pers. comm.) noted discrepancies of 10% by weight in the mass balance for samples of Vertisols (Vertosols; Isbell 1996) on the Darling Downs,

Queensland. In such soils, derived from basalt, the smectite present may contain up to 20% Fe_2O_3 , e.g. Pelican clay (Norrish and Pickering 1983). Its density calculated from the composition of the unit cell assuming a basal spacing of 9.6 Å would be $>3.0 \text{ mg/mm}^3$, which would account for the discrepancy.

Possible practical applications

Combining measurements of settling velocity with particle density could be of particular interest where particles are already dispersed. For example, the turbidity of waters is due to particles peptised by the organic matter in eroded surface soil or by compounds produced by microbial action in the water. It would be interesting to relate density to settlement velocity and carbon content. For this, fine fractions could be separated by centrifuging and a much smaller plummet used. A second example is in dams filled with mine tailings. There, the particles are dispersed, but not necessarily peptised. The particles usually have a wider range of density than in soils, partly because iron oxide is present as separate particles rather than bonded to the clay as in red-brown earth subsoils (Greenland *et al.* 1968; Emerson 1983). Also tailings are usually discharged at the edge of the dam, the finer and lighter particles then flowing out to the centre. It could be useful for reclamation purposes to know the accompanying changes in fraction density and therefore composition as well as allowing more accurate calculation of actual particle sizes using Stokes Law.

A quicker way of measuring a weight percentages of particles in suspension is by the attenuation of a collimated beam of X-rays. The difficulty is that the values obtained are strongly influenced by the Fe content of the particles (Buchan *et al.* 1993).

Conclusions

Measurements on separated silt and clay fractions show that differences in density need to be determined to 0.01 mg/mm^3 to be really useful in detecting variations in fraction composition. This sets the precision with which both the weight concentration of a fraction in a soil suspension and the density of the suspension should be measured in order to deduce fraction density. If, in a 2% suspension, about 50% of the particles by weight are in one fraction, then the precise concentration of the particles in this fraction can be measured by taking a 5-mL aliquot, drying, and weighing. However, the precision with which percentages can be measured using the plummet balance of Marshall (1956) is not quite good enough. Precision should be improved if the upthrust on the plummet were measured directly using a 4-place balance. Such a change would also overcome the difficult process of calibrating the balance to the required accuracy and also enable sufficiently accurate values to be obtained with a shorter plummet. In turn, this could help to deduce accurate densities in centrifuged suspensions. Ideally, to ensure smooth reproducible entry of the plummet, the suspension would be moved up and down automatically while the plummet remains fixed.

It may be possible to associate the density of fractions with particular soil orders. For example, a density of about 2.84 mg/mm^3 seems to be common to the Na-form of the clay fractions of red-brown earths in Southern Australia.

Acknowledgments

Thanks are due to M. Raven, CSIRO Land and Water, Adelaide for analysing the silt fraction, Dr R. Bryant, University of Swansea, for supplying the kaolinite, H. J. Beatty, formerly of CSIRO Division of Soils, Canberra for helpful comments on using the plummet balance, and to the referees for useful comments and suggestions.

References

- Buchan GD, Grewal KS, Claydon JJ, McPherson RJ (1993) A comparison of Sedigraph and pipette methods for soil particle-size analysis. *Australian Journal of Soil Research* **31**, 407–417.
- Burford JR, Greenland DJ, Quirk JP (1962) Surface area determinations and the clay mineralogy of a Red Brown Earth. *Nature*, 361–12.
- Day PR (1953). Experimental confirmation of hydrometer theory. *Soil Science* **75**, 181–86.
- Deshpande TL, Greenland DJ, Quirk JP (1968). Changes in soil properties associated with the removal of iron and aluminium oxides. *Journal of Soil Science* **19**, 108–122.
- Emerson WW (1964) The slaking of soil crumbs as influenced by clay mineral composition. *Australian Journal of Soil Research* **2**, 211–217.
- Emerson WW (1983) Inter-particle bonding. In 'Soils: an Australian viewpoint'. pp. 477–498. (CSIRO Australia/Academic Press: London)
- Emerson WW, Self PG (1994). Neutralized tailings and sulphates: evaluating bulk properties. *Geotechnique* **44**, 495–501.
- Greenland DJ, Oades JM, Sherwin TW (1968). Electron-microscope observations of iron oxides in some red soils. *Journal of Soil Science* **19**, 123–126.
- Huang PM (1977) Feldspars, olivines, pyroxenes and amphiboles. In 'Minerals in soil environments'. (Ed. RC Dinauer) pp. 553–602. (SSSA: Madison, WI)
- Hutton JT (1955) A method of particle size analysis of soils. Divisional Report No. 11, CSIRO Division of Soils, Adelaide.
- ICT (1932) International Critical Tables, III, 79. (National Resources Council, USA)
- ISSS (1934) Trans. 1st Comm. International Society of Soil Science, pp. 303–304, Versailles.
- Isbell RF (1996) 'The Australian soil classification'. (CSIRO Publishing: Melbourne)
- Lambe TW (1951) 'Soil testing for engineers. Ch. II. Specific gravity test.' pp. 22–31. (John Wiley & Sons: New York)
- Litchfield WH (1951) Soil Survey of the Waite Agricultural Research Institute. Divisional Report 2/51, CSIRO, Adelaide.
- McIntyre DS (1974) Definitions of particle size by settling velocity and standard sieves. In 'Methods for analysis of irrigated soils'. (Ed J Loveday) pp. 188–92. (CAB and CSIRO)
- McIntyre DS, Loveday J (1974) Particle-size analysis. In 'Methods for analysis of irrigated soils'. (Ed. J Loveday) pp.78–87. (CAB and CSIRO)
- Marshall TJ (1956) A plummet balance for measuring the size distribution of soil particles. *Australian Journal of Applied Science* **7**, 142–147.
- Merry RH (1985) A summary listing of the archival soils collection of the Division of Soils. Divisional Report No. 80. CSIRO, Adelaide.
- Norrish K, Pickering JG (1983). Clay Minerals. In 'Soils: an Australian viewpoint'. Division of Soils, CSIRO. pp. 281–308 (CSIRO: Melbourne/Academic Press: London)
- Radoslovich EW (1958) Clay mineralogy of some Australian red-brown earths. *Journal of Soil Science* **9**, 242–251.
- Sheldrick BK, McKeague JA (1976). Tests of the plummet balance and a comparison of some procedures for particle size analysis. *Canadian Journal of Soil Science* **56**, 55–57.
- Slager S, Koenigs FFR (1964) Particle size analysis of soils by means of a torsion balance and a plummet. *Sedimentology* **3**, 240–52.

Appendix

The density (ρ) of an aqueous suspension containing m mg/mm³ of particles with an average dry density of ρ_s mg/mm³ is given by:

$$\rho = \rho_w + m(1 - \rho_w/\rho_s) \quad (1)$$

where ρ_w is the density of water.

If a plummet balance is calibrated for particles of density ρ_c , p_i is the net percentage by weight indicated on the scale of the balance after subtracting the blank and p is the true percentage (by pipette method), then putting $\rho_w = 1.0$, from Eqn 1:

$$p_i(1 - 1/\rho_c) = p(1 - 1/\rho_s) \quad (2)$$

At constant temperature, to determine the effect of a small error in m on ρ_s , differentiating Eqn 1 gives:

$$(1/m).dm/d\rho_s = 1/\rho_s(\rho_s - 1) \quad (3)$$

Ideally, a difference in particle density of 0.01 mg/mm³ needs to be detected. So for a 2% suspension of a soil containing 50% clay (by pipette), using Eqn 2, if the balance is calibrated for a ρ_s of 2.60 mg/mm³, it has to be possible to distinguish between readings of 50.1% and 50%.

Eqn 1 does not strictly apply since the suspension medium is not water, but the dispersant solution. From Eqn 1, treating ρ_w as the variable:

$$dm/m = (1/\rho_s).d\rho_w \quad (4)$$

The balance was calibrated so that 100% corresponds to a density difference of 0.0123 mg/mm³. So the blank reading of 6.5% corresponds to a density difference in the 'water' of 0.0008 mg/mm³. From Eqn 3 the resultant percentage error in m is negligible. Note that a change in temperature of 1°C has about 3 times the effect.

Manuscript received 27 October 1997, accepted 13 October 1999